Fluent BioSciences

Single Cell Biology Made Simple

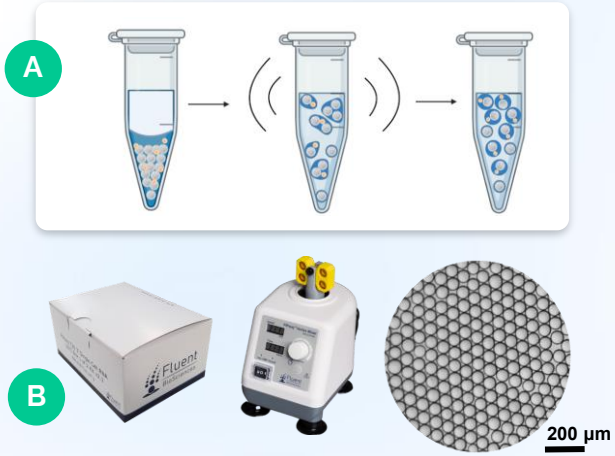
PIPseq Eliminates Limitations in Single Cell Analysis

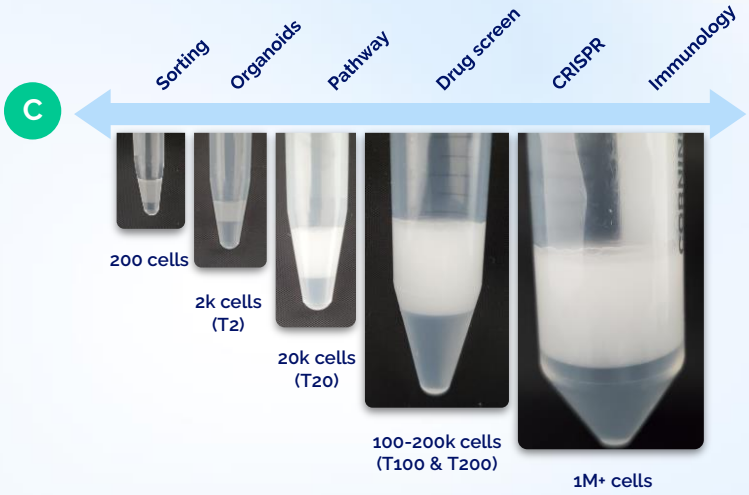
- Expensive instrumentation and consumables
- Limited scale
- Rigid workflows
- Significant expertise required

PIPseq eliminates these barriers, making single cell analysis available to any researcher



PIPseq Technology



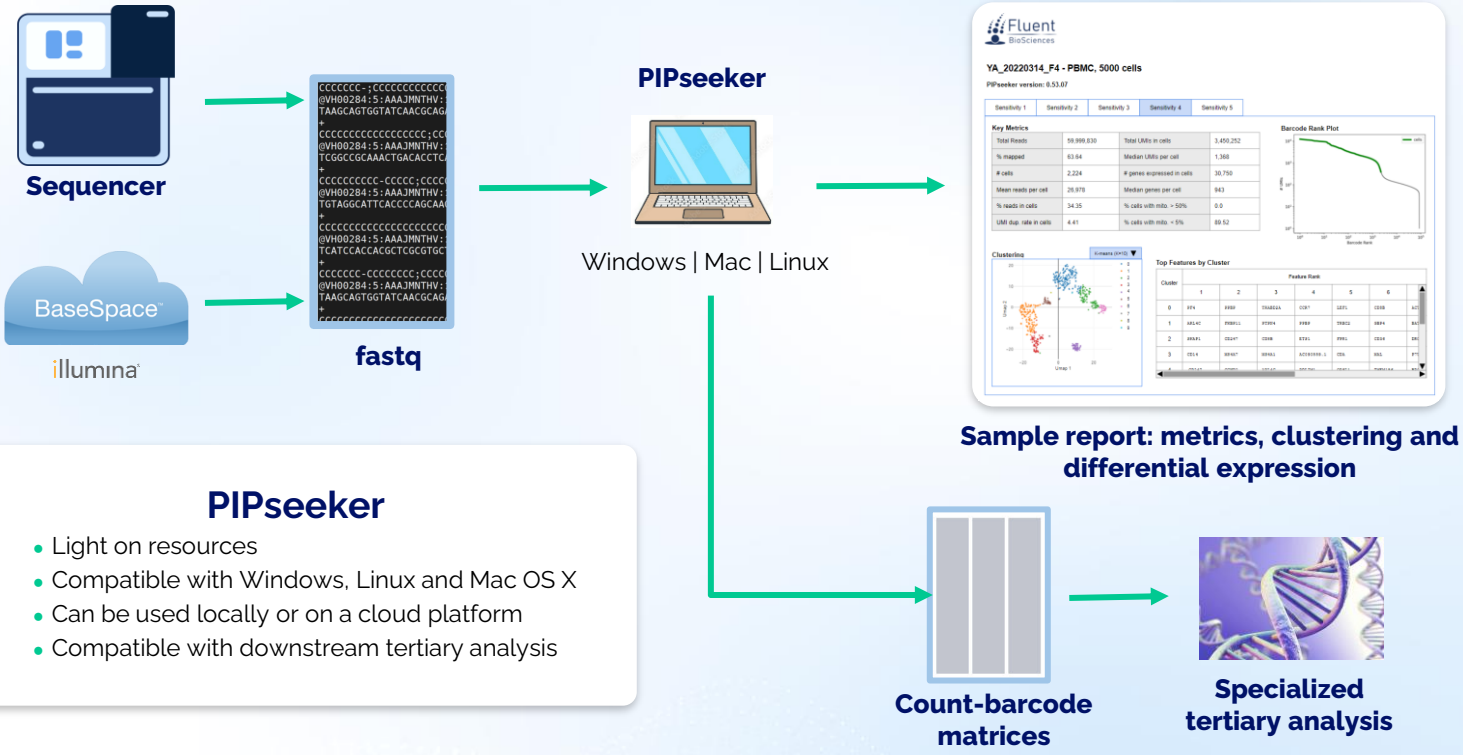


A— Single-cell capture in minutes without complex instrumentation

B— Reagent kit with low-cost equipment enables generation of uniform nanoliter-scale reactions

C— Scalable reactions from hundreds to millions of cells

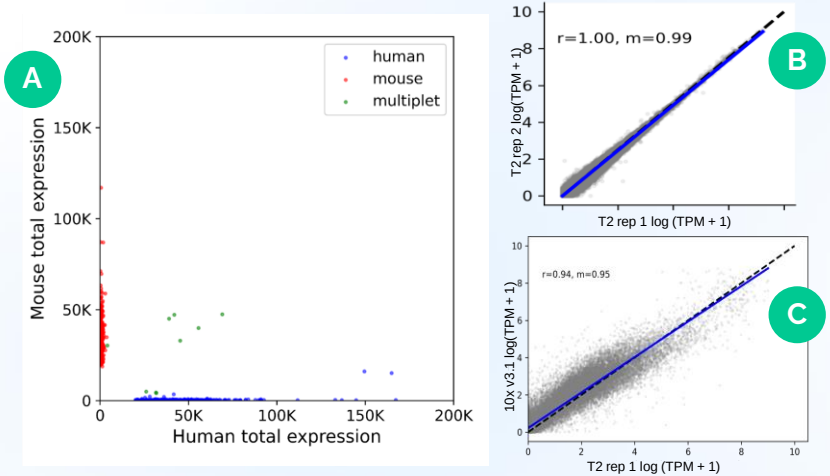
PIPseeker: Fluent's scRNAseq Analysis Pipeline



PIPseeker

- Light on resources
- Compatible with Windows, Linux and Mac OS X
- Can be used locally or on a cloud platform
- Compatible with downstream tertiary analysis

PIPseq Performance and Specifications



A— Human / mouse (HEK / 3T3) cell mixture separation shows very low multiplet rate.

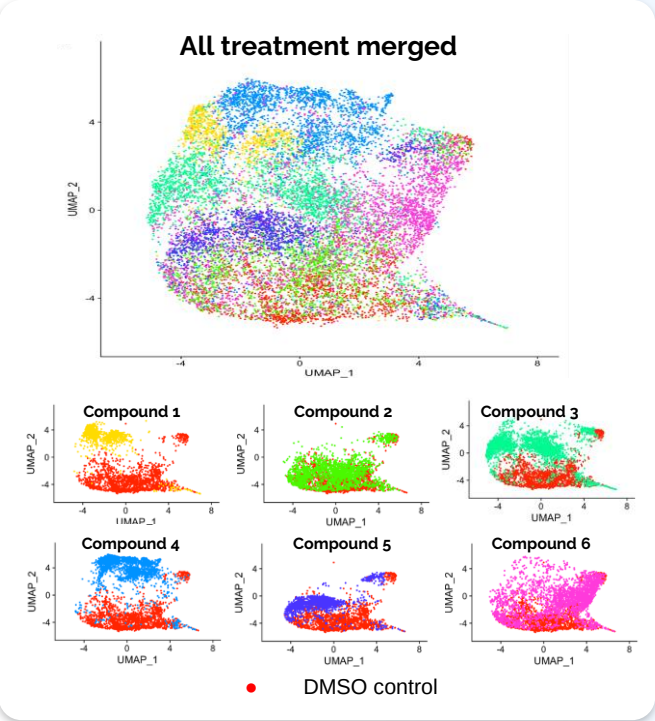
B— Very high reproducibility of detected genes per cell in replicate PIPseq

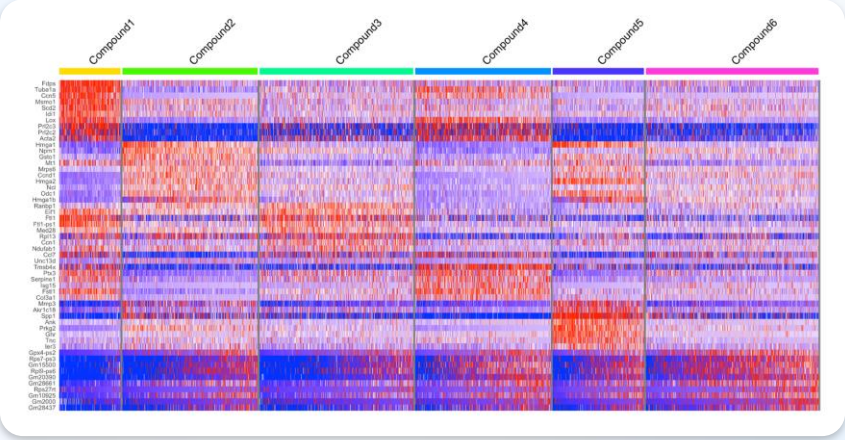
C— High concordance of detected genes with other scRNAseq techniques

Kit Format	T2	T20	T100
Cell output (per reaction)	Up to 2,000 cells	Up to 20,000 cells	Up to 100,000 cells
Example Use Cases	• Cell culture • Flow sorting • Drug testing	• Complex tissue • Clinical sample variation • Cell Atlas	• Neuroscience • Immune cell • Multiplexed samples
Reactions per kit	8	4	2
Cell capture efficiency	>40%		
Multiplet rate	<5%		
Time to capture	10 min hands-on		
Time to library	2 days		
Specifications*	Genes/cell: >2,000; Transcripts/cell: >7,000 (at 20K reads per called cell)		

* Validated with HEK/3T3 human-mouse cell standards.

PIPseq T2: Multivariable Drug Dosing in Glioblastoma Xenografts



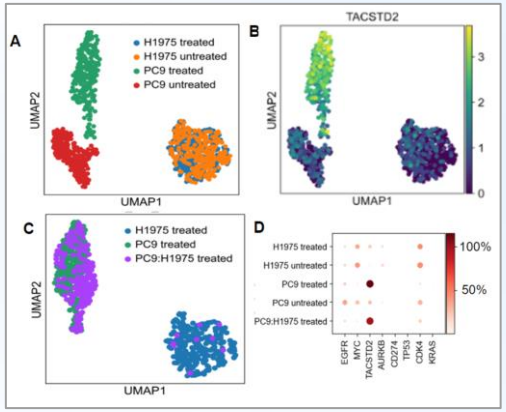


- Mice with xenograft glioblastoma treated with control or 1 of 6 test therapeutics
- Tumors excised and processed by PIPseq T2
- Differential gene expression analysis reveals mechanism of action

PIPseq enables large-scale evaluation of treatment conditions at single cell resolution

PIPseq T2: Experimental Flexibility in Multi-Factor Studies

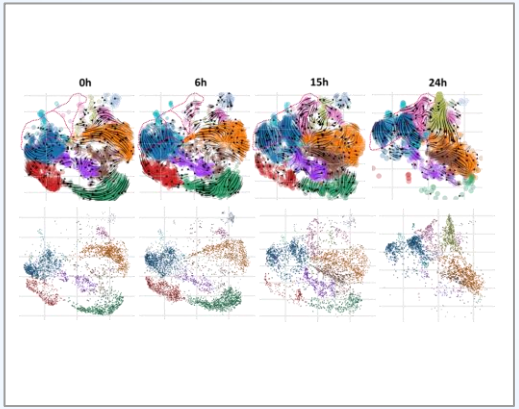
Drug Discovery



HARVARD UNIVERSITY

Single-cell response to therapeutic intervention

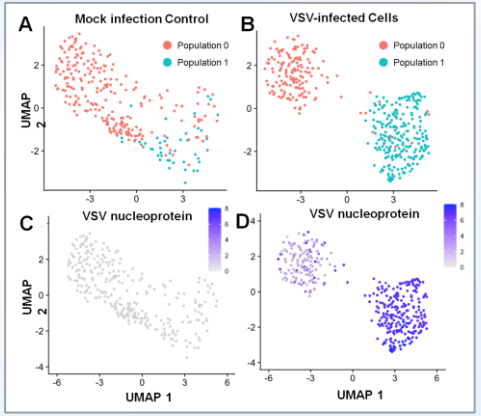
Multi-sample time course studies



RUTGERS

Time course of therapeutic action in intestinal organoid model

Disease Monitoring

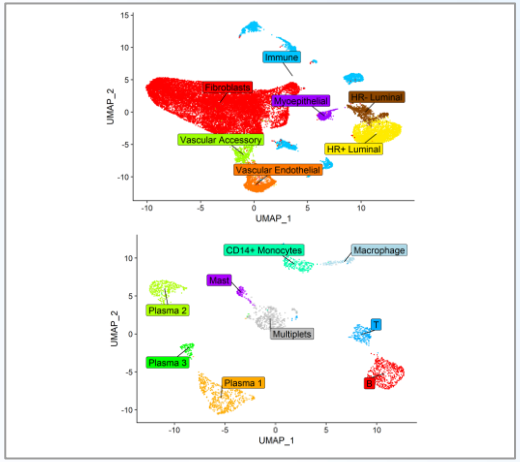


BOSTON UNIVERSITY

Host response to viral infection

PIPseq T20: Cell Diversity in Complex Samples

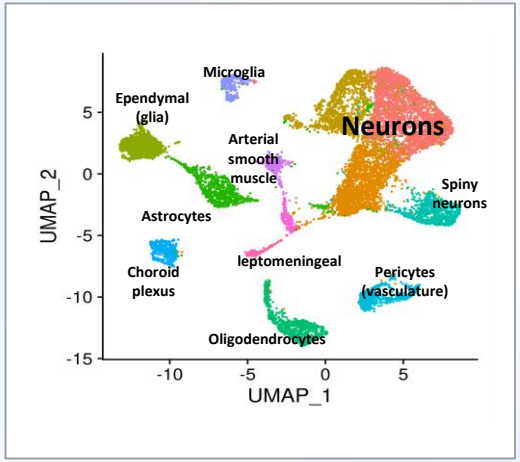
Clinical Tissue Profiling



University of California San Francisco

Rare cell identification from human breast tissue

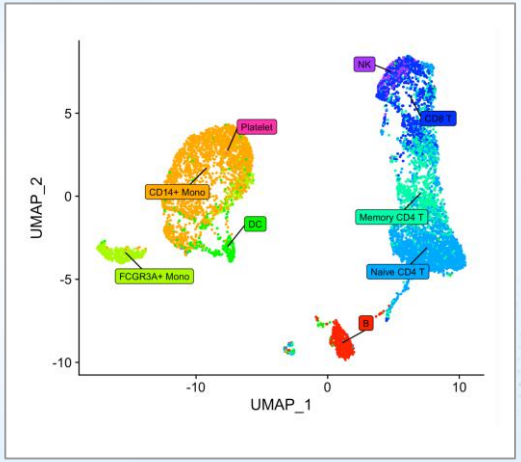
Neuroscience



NYU

Fresh dissociated mouse brain

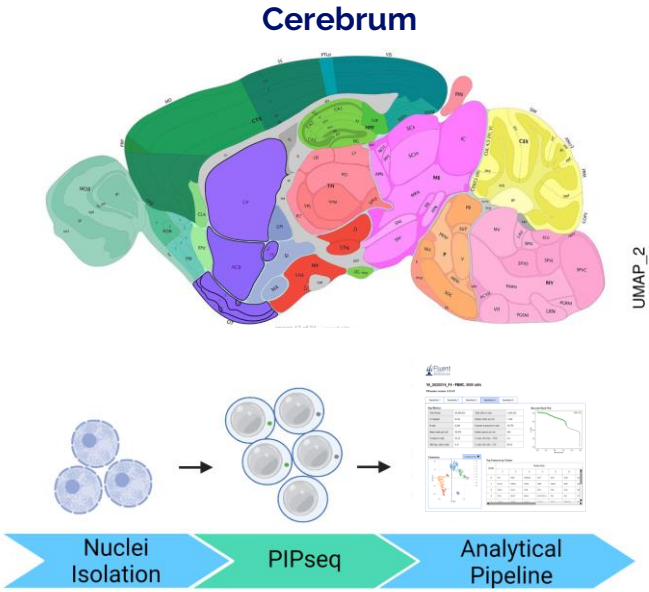
Immunology

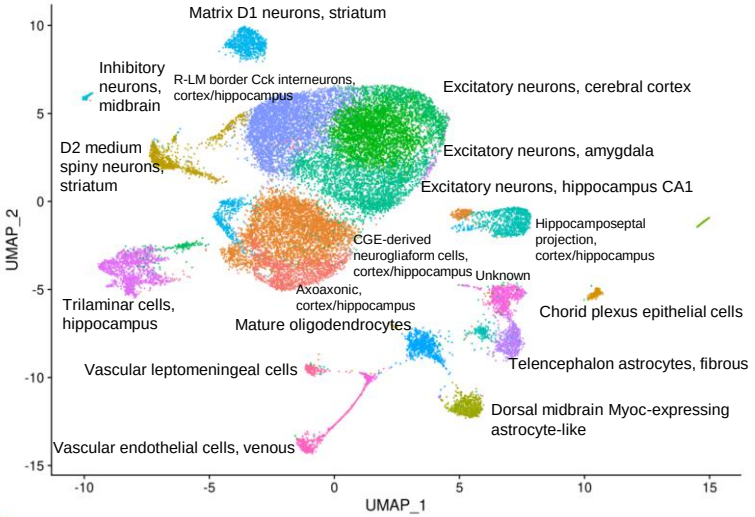


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Peripheral blood mononuclear cells

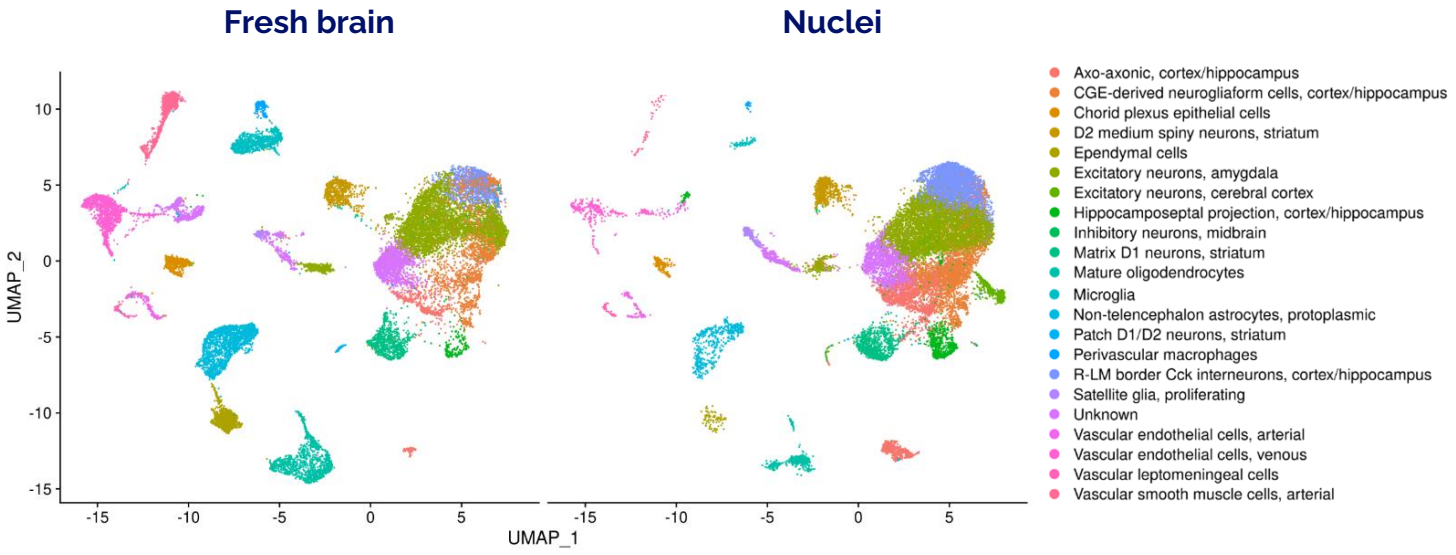
PIPseq T20: Nuclei from Whole Mouse Brain





Simple Nuclei EZPrep™ isolation yields highly resolved brain populations

PIPseq T20: Mouse Whole Brain Cells and Nuclei

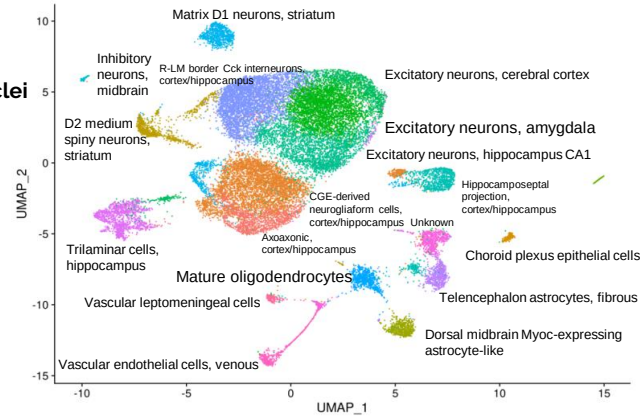


Nuclei capture the full diversity of cell types recovered from fresh brain tissue

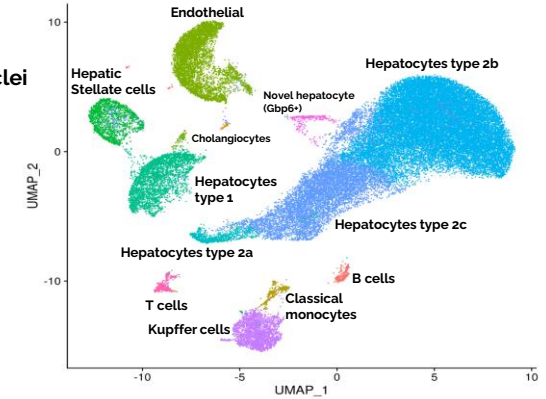
PIPseq is compatible with nuclei prepared from frozen tissue

Fluent BioSciences

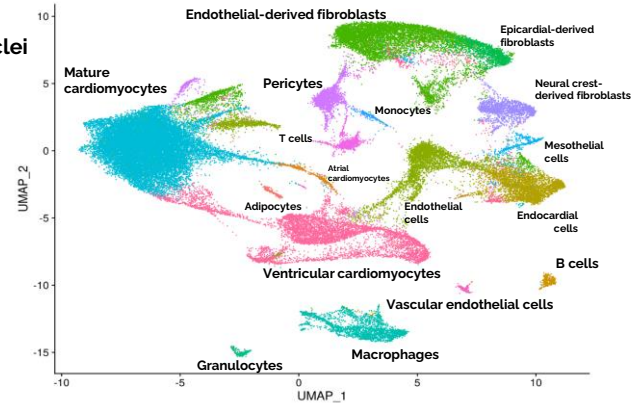
Brain
26k nuclei



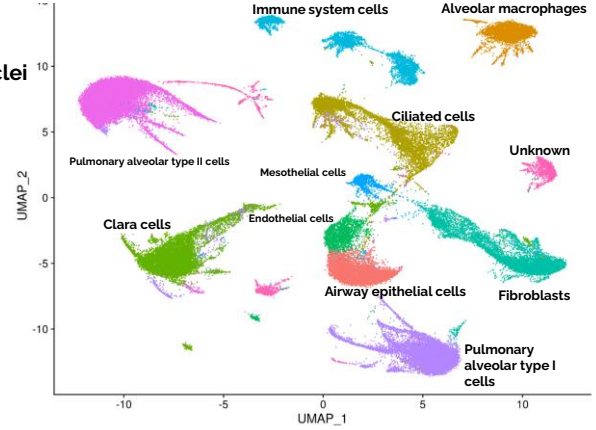
Liver
47k nuclei



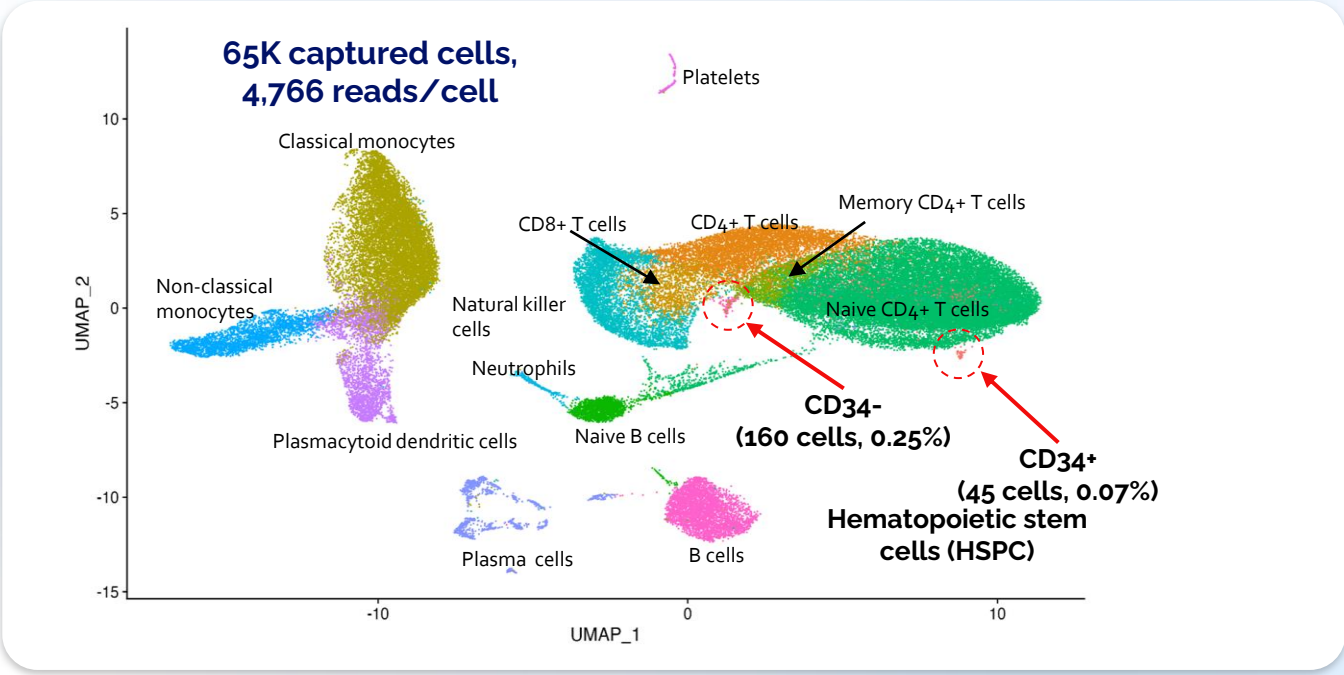
Heart
58k nuclei



Lung
68k nuclei



PIPseq T100: Immune Cell Diversity in PBMCs

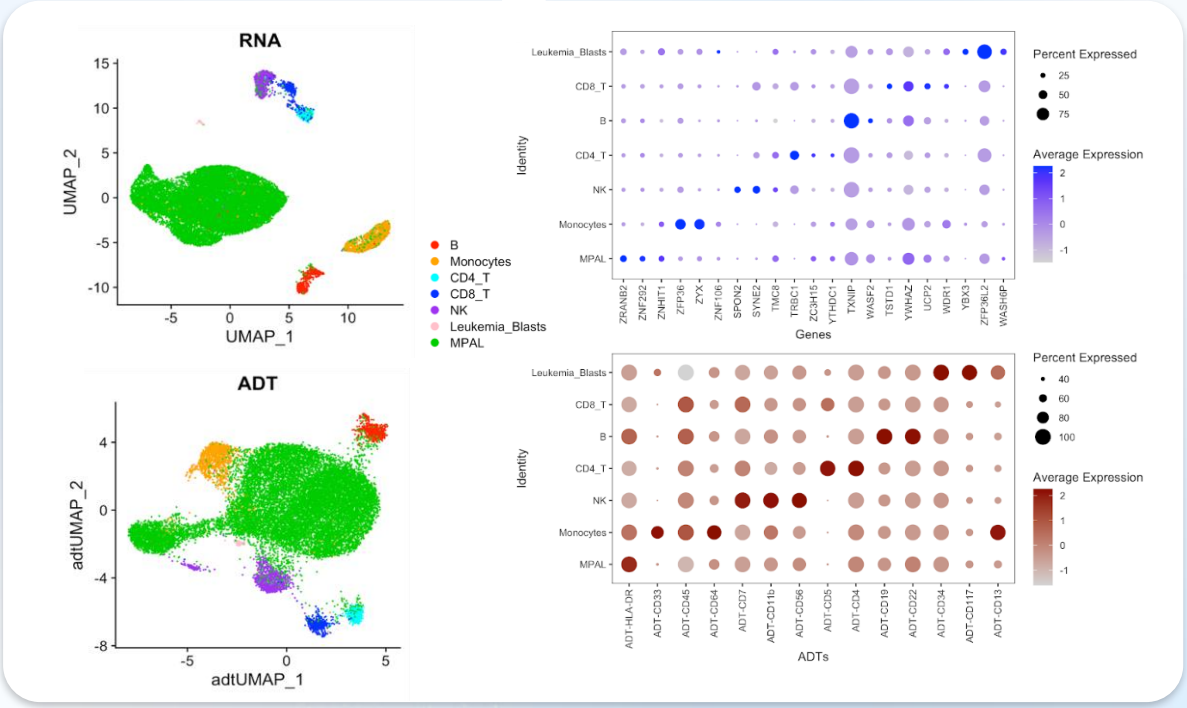


PIPseq resolves rare PBMC subpopulations with low depth sequencing

Multiplex Protein Epitope Labeling in PIPseq

- Fluent has partnered with BioLegend to enable protein epitope labeling and cell hashing in PIPseq
- Antibody-derived tags (ADTs) provide rich multi-omic detail of cellular protein expression in the context of 3' transcriptome sequencing
- Here, a 19 ADT panel was used to differentiate cell populations in Mixed Phenotype Acute Leukemia (MPAL)

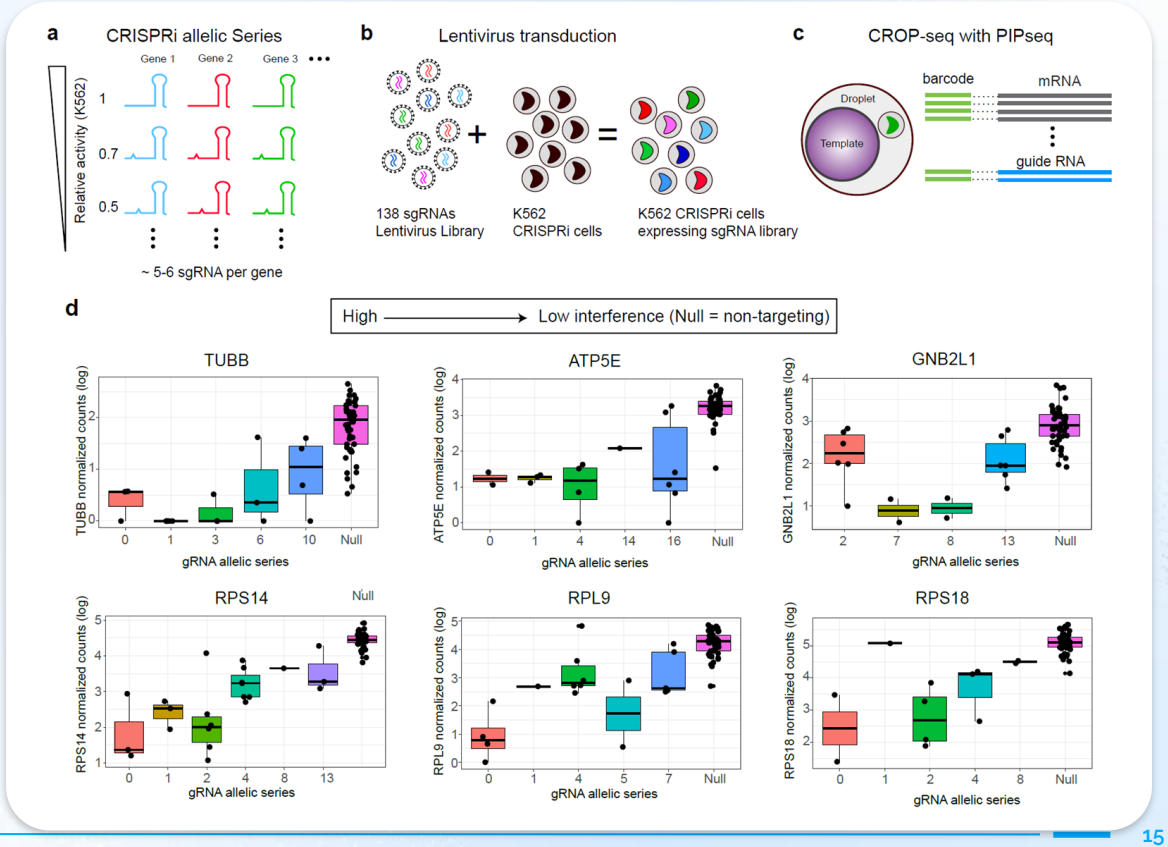
PIPseq enables multi-omic analysis



CRISPR screening in PIPseq

- Preliminary experiments demonstrate CRISPR screening in PIPseq
- 138 guide RNA library
- CrOP-seq - like protocol
- Correlation of transcriptomic knockdown with activity of CRISPR guide RNA

PIPseq enables highly-scalable perturbation screening



Products & Pricing

Option 1: Kit Purchase

	Cell Output (per rxn)	Reactions per Kit	Kit Price
PIPseq T2	Up to 2,000	8	$2,400 ($300/sample)
PIPseq T20	Up to 20,000	4	$3,600 ($900/sample)
PIPseq T100	Up to 100,000	2	$6,400 ($3,200/sample)
Epitope Seq	User defined	8	$400 ($50/sample)
UDI-96 Kit	User defined	96	$800 ($8.33/sample)





Starter kit (one-time): $2,500
Includes vortexer, dry bath, platefuge and accessories

Contact: sales@fluentbio.com

Option 2: Fluent Services

User Input	Prepare cell suspension, capture cells with Fluent kit and ship
Fluent Workflow	Generate cDNA, prepare sequencing libraries, perform NGS and primary data analysis
Sample Types	Dissociated cells from solid tissue or cultured cells (< 30 µm)
Turnaround Time	< 4 weeks
Price	Contact sales@fluentbio.com to quote your project

Contact us to learn how to become a validated PIPseq service provider

PIPseq: Enabling Single Cell in Every Laboratory

Fluent's flexible, cost effective platform and expanding portfolio of applications providers researchers the tools necessary for ambitious science

	Fluent BioSciences
NO CAPITAL INVESTMENT	✓
COST EFFECTIVE	✓
ACCESSIBILITY	✓
SCALE (PER RXN) 100 TO >1M CELLS	✓
SAMPLE TYPE (FRESH + FIXED + NUCLEI)	✓
MULTI-OMICS CAPABILITY	✓
POSITIONED FOR THE CLINIC	✓



Acknowledgements

Advisors and Collaborators

Adam Abate, Ph.D.
Professor, Bioengineering
Fluent Biosciences Co-founder

Zev Gartner, Ph.D.
Professor, Pharmaceutical Chemistry

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Assistant Professor, Bioengineering
Fluent Biosciences Advisor

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Shane Liddelow Ph.D.
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SBIR Support

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1 R44 CA 244049-01

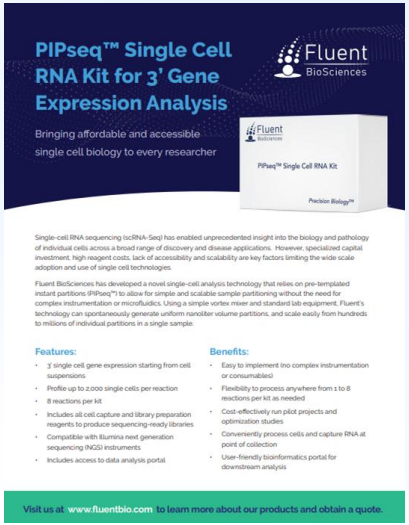
NIH National Institute of General Medical Sciences
1 R43 GM 137648-01
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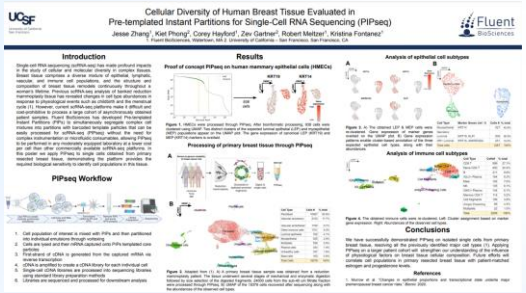
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